

Anchoring Large Language Models to Empirical Behavioral Evidence: A Hybrid Approach for Vaccination Uptake Simulation

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Abstract

Objective: To develop and validate *empirical-frontier regularization (EFR)*, an inference-time procedure anchoring LLM outputs to a DCE-derived empirical frontier without model retraining.

Methods: A Wuhan vaccination DCE ($N=1,027$) was combined with LLM simulation on a 64-state policy grid. EFR computes $\pi_{\text{EFR}} = \lambda \bar{\pi}_{\text{LLM}} + (1 - \lambda) P_{\text{emp}}$ with $\lambda=0.25$ fixed as a conservative anchoring design choice, not a test-set-optimized predictive optimum; held-out log loss and Brier are reported as sensitivity diagnostics rather than as the criterion used to select λ .

Results: On the held-out DCE subset, Static-BDT (EFR) achieved log loss 0.699 and Brier 0.252 versus Pure-DCE (0.679 / 0.242) and unconstrained LLM (0.827 / 0.314). The unconstrained LLM violated empirical waiting-time monotonicity on 62.5% (Qwen2.5-72B-Instruct), 34.4% (DeepSeek V4), and 10.4% (MiroThinker-1.7-mini) of the 64-state scenarios; after EFR ($\lambda = 0.25$) the same wait-axis MVR-Wait dropped to 30.2%, 7.3%, and 8.3% respectively. Frontier-alignment rank correlation ρ for the same three models rose from 0.061 / 0.302 / 0.531 (raw LLM) to 0.689 / 0.812 / 0.756 (EFR) on the 64-state grid, all exceeding the construction-only null distribution ($\rho=0.367$, 95% CI [0.147, 0.566]).

Limitations: Validation used a single Wuhan DCE, and the out-of-design narrative scenarios lacked human ground truth; generalizability and out-of-design behavioral validity require external replication.

Conclusion: EFR extends an existing DCE with constrained narrative-policy evaluation by assigning most of the weight to the empirical reference; the fixed $\lambda = 0.25$ anchor bounds, rather than eliminates, the deviation of LLM outputs from the DCE-derived frontier.

Keywords: empirical-frontier regularization; large language models; discrete choice experiment; health technology assessment; behavioral simulation

1 Introduction

Large language models (LLMs) are increasingly used as simulated decision-makers for healthcare policy [18, 17, 26]. Their appeal is that they can interpret rich natural-language scenarios that conventional structured decision models cannot. However, linguistic fluency does not guarantee behavioral reliability: an unconstrained LLM may assign higher acceptance to a longer-waiting policy than to an otherwise identical shorter-waiting policy, contradicting empirically observed preference patterns and producing unreliable policy rankings.

Current approaches do not explicitly enforce empirically validated behavioral constraints [1, 11, 19]: LLM agents may produce decisions that appear linguistically plausible while violating basic preference relationships implied by observed human behavior.

One source of such constraints is the discrete choice experiment (DCE), which estimates how observed human preferences vary with decision attributes [5]. In this study, we treat DCEs not as models to be replaced by LLMs, but as sources of empirically validated behavioral knowledge. A DCE-estimated choice frontier defines the preference ordering and directional relationships supported by observed responses. The challenge is to combine this empirical behavioral knowledge with the representational flexibility of LLMs, so that simulated predictions can incorporate an LLM-derived narrative component while remaining closer to DCE-implied structure; whether the resulting narrative extension is behaviorally more accurate on out-of-design scenarios is not established by the present study, because no human ground truth is available outside the original DCE design space.

We propose a **hybrid approach** that anchors LLM-generated choice probabilities to empirical behavioral evidence from DCEs. The method estimates an empirical choice frontier from DCE responses and blends the LLM’s raw probability with the frontier through a transparent weighted-averaging procedure. The adjustment is auditable: the raw LLM probability, the empirical frontier value, the blending weight, and the final adjusted probability are all directly inspectable. The method requires no retraining. The same LLM can be anchored using different empirical datasets, enabling domain-specific grounding while preserving general capabilities.

We evaluate the approach using a real-world vaccination-timing DCE involving 1,027 adults in Wuhan, China. This study makes three contributions:

1. We introduce a lightweight inference-time adjustment procedure for grounding LLM-based simulation in empirically estimated human behavioral knowledge.
2. We show that the proposed adjustment improves behavioral reliability across multiple LLMs, reducing behavioral inconsistencies, improving agreement with the empirical choice frontier, and improving probability calibration relative to unconstrained LLM agents.
3. We demonstrate that empirically grounded behavioral models can be extended to richer natural-language healthcare policy scenarios while preserving auditable behavioral constraints.

2 Related Work

Our work connects two research areas: LLM-based synthetic agents for healthcare simulation and empirical behavioral models derived from discrete choice experiments (DCEs). While these two directions have developed largely independently, both seek to improve the realism and reliability of decision support. Our work lies at their intersection by incorporating empirically estimated behavioral knowledge into LLM-based synthetic agents through inference-time regularization.

Prior LLM-for-HTA work in this journal. Recent issues of *Value in Health* have featured LLMs along trajectories distinct from the behavioral-simulation axis we pursue: evidence extraction [21], sponsor-HTA conversation simulation [16], and microsimulation [23]. We deploy LLMs as narrative-extension mechanisms anchored to a DCE-derived empirical frontier, not as tools for review or negotiation. The closest analogue outside HTA is Twin-2K-500 [24], which validates LLM-as-consumer simulation in marketing; we extend this line to health policy. A pre-existing LLM-powered social digital twin framework [13] is conceptually adjacent but does not anchor to a pre-collected empirical choice frontier.

2.1 LLM-Based Synthetic Agents

Large language models (LLMs) have enabled a new generation of synthetic agents capable of generating natural-language reasoning and simulating human decision making across diverse domains [3, 18, 17]. In healthcare, these agents have been explored for public health communication, policy evaluation, and behavioral simulation [20, 6]. Their principal advantage is representational flexibility: unlike conventional statistical models, LLMs can process rich textual descriptions and respond to scenarios that extend beyond predefined structured variables.

However, behavioral realism remains a major challenge: LLM responses may violate empirically established relationships or produce implausible preference orderings [1, 11, 10]. Prompt engineering can reduce some inconsistencies but provides no explicit mechanism for enforcing empirically grounded behavioral constraints [28, 27]; the mechanism cannot guarantee compliance across diverse narrative phrasings outside the original prompt template. We adopt the LLM-as-narrative-component framing of Argyle et al. [2], Horton [11]: the LLM’s narrative output is one input to a convex blend with an empirical choice frontier, not a stand-alone behavioral model.

2.2 Empirical Behavioral Models from Discrete Choice Experiments

Discrete choice experiments (DCEs) provide a well-established framework for estimating how individuals trade off policy-relevant attributes such as waiting time, vaccine efficacy, side effects, and financial incentives [22, 5]. Under the random utility framework [25, 15], DCE responses can be used to estimate an empirical choice frontier that preserves interpretable relationships between decision attributes and observed choices. Mixed-logit models further account for preference heterogeneity across respondents [8, 9].

In this study, we treat the DCE frontier as an empirically estimated source of behavioral knowledge rather than as assumption-free ground truth. While empirical choice models provide transparent and behaviorally interpretable predictions within their original experimental design, they cannot directly operate on rich natural-language policy scenarios.

2.3 Bridging Empirical Behavioral Models and LLM-Based Agents

Existing LLM-based synthetic agents provide representational flexibility but lack mechanisms for enforcing empirically validated behavioral constraints [14, 23]. DCE-derived behavioral models provide transparent and interpretable structure but remain confined to their experimental design. Existing approaches trade flexibility for reliability or vice versa.

3 Methods

This study integrated two methodological components: (1) a discrete choice experiment to estimate empirical vaccination preferences, and (2) a procedure for aligning LLM predictions with these empirical patterns. The following sections describe each component and the validation approach.

3.1 Overview

Section 3.2 describes the DCE design and empirical frontier; Section 3.3 presents the EFR framework. Section 4.2 reports the 64-state evaluation, and Section 4.3 reports the out-of-design narrative evaluation; Section 5.6 concludes.

3.2 Stage 1: Empirical Behavioral Knowledge Extraction

The first stage extracts empirical behavioral knowledge from a vaccination-timing DCE conducted among 1,027 adults in Wuhan, China. This knowledge is an empirical choice frontier that serves as the behavioral reference for inference-time regularization.

Stakeholder involvement and instrument design. Two physician interviews and two rounds of twelve cognitive interviews (ISPOR good-research-practices [12, 4]) refined the choice-task framing.

Construction of the Empirical Behavioral Reference. The empirical behavioral reference was constructed from a 6-parameter grouped/conditional logit with five attribute coefficients and an opt-out alternative-specific constant, estimated on the Wuhan DCE. The frontier maps each candidate policy to a probability of acceptance, providing a deterministic signal that anchors the LLM’s narrative choices.

3.3 Stage 2: Inference-Time Adjustment Procedure

The adjustment algorithm aligns LLM outputs with the empirical reference by computing $\pi_{\text{adj}}(y = 1 \mid \mathbf{x}_g) = \lambda \cdot \pi_{\text{LLM}}(y = 1 \mid \mathbf{x}_g) + (1 - \lambda) \cdot P_{\text{emp}}(y = 1 \mid \mathbf{x}_g)$ (the EFR equation $\pi_{\text{adj}} = \lambda\pi_{\text{LLM}} + (1 - \lambda)P_{\text{emp}}$; full derivation in Appendix A.1 of the Supplementary Material).

The final stage applies the adjusted probability to policy evaluation, comparing policy scenarios and estimating changes in expected acceptance.

For a baseline scenario $\mathbf{x}$ and an intervention that changes attribute x_k to x'_k , the policy response is:

$$\Delta_k(\mathbf{x}) = \pi_{\text{adj}}(y = 1 \mid \mathbf{x}') - \pi_{\text{adj}}(y = 1 \mid \mathbf{x}),$$

where $\mathbf{x}'$ denotes the post-intervention scenario. Averaging over comparable scenarios gives the mean policy response for attribute k .

Because evaluation operates only on the adjusted probability surface, the approach can be deployed locally without transmitting individual-level DCE responses to external providers. The resulting policy-response quantities are descriptive summaries and should not be interpreted as causal treatment effects.

Table 1: Behavioral consistency with the empirical reference before and after EFR on the 64-state grid. The empirical frontier is the corrected 6-parameter grouped/conditional logit on the full DCE sample. Spearman’s ρ is a diagnostic only; the held-out DCE log loss (Table S6 of the Supplementary Material) is the primary benchmark.

Model	Method	MSE $\downarrow$	MAE $\downarrow$	MVR-Wait (%) $\downarrow$	Spearman’s ρ $\uparrow$
Qwen2.5-72B-Instruct	Raw LLM	0.0258	0.1269	62.5	+0.061
	EFR	0.0016	0.0317	30.2	0.689
DeepSeek V4	Raw LLM	0.0379	0.1570	34.4	0.302
	EFR	0.0024	0.0392	7.3	0.812
MiroThinker-1.7-mini	Raw LLM	0.0872	0.2644	10.4	0.531
	EFR	0.0054	0.0661	8.3	0.756

4 Experimental Evaluation

4.1 Experimental Setup

We evaluated EFR using three open-source LLMs (Qwen2.5-72B-Instruct, DeepSeek V4, MiroThinker-1.7-mini) via their respective public inference endpoints (Hugging Face for Qwen2.5-72B-Instruct; provider endpoints for DeepSeek V4 and MiroThinker-1.7-mini; access dates and endpoints are itemized in Table S1 of the Supplementary Material and the supplementary archive). Decoding parameters were matched across models where the API permitted, with Qwen2.5-72B-Instruct and DeepSeek V4 sampled at temperature = 0.7, top_p = 1.0, max_new_tokens = 512, and $R = 10$ repeated generations per state to average out sampling noise; MiroThinker-1.7-mini was sampled at temperature = 0.0 (deterministic), top_p = 1.0, max_new_tokens = 256, and $R = 10$ repetitions. The empirical choice frontier was estimated from a corrected 6-parameter grouped/conditional logit with five attribute coefficients and an opt-out alternative-specific constant on the full Wuhan DCE sample ($N = 1,027$ respondents) on vaccination timing (ISPOR good-research-practices [12, 7]). All LLM evaluations in this paper comply with the ViH/ISPOR ELEVATE-GenAI reporting framework; ELEVATE-GenAI reporting details are provided in Appendix C of the Supplementary Material.

4.2 Behavioral Consistency with the Empirical Frontier

Table 1 summarizes behavioral consistency on the 64-state evaluation grid (wait $\in \{0, 2, 4, 6\}$ months, efficacy $\in \{0.3, 0.5, 0.7, 0.9\}$, side-effect burden level $\in \{0, 1, 2, 3\}$, cash and origin held at grid reference values). MVR is the wait-axis monotonicity violation rate (fraction of (eff, se)-matched state pairs in which longer wait is paired with higher π than shorter wait); ρ is Spearman’s rank correlation between π and P_{emp} across the 64 states. A complementary task-level analysis on the LLM-matched A-versus-B subset of the held-out 20% respondent split ($n_{\text{tasks, matched}} = 78$ under the canonical 11-of-12-cell populated intersection) yields the same qualitative ordering: Pure-DCE task-level log loss 0.3307, Static-BDT ($\lambda = 0.25$) 0.4307, and unconstrained LLM 0.8521 (Table S15). This is a complementary task-level A-versus-B validation (the canonical full three-alternative matched subset is empty; the held-out sample admits only two-alternative non-opt-out tasks under the 11-of-12-cell populated intersection), and the fact that the task-level ordering matches the alternative-level held-out ordering reported in Figure 1 and Table S6 (and the same ordering also appears on the held-out alt-level binary 813-row matched subset, Pure-DCE 0.6789 vs. EFR 0.6985).

vs. LLM 0.8273) corroborates that the alt-level binary scoring is not an artefact of the score choice. EFR ($\lambda = 0.25$, the conservative anchoring design choice fixed across all three models) reduced MSE by 94% (Qwen2.5-72B-Instruct: $0.0258 \rightarrow 0.0016$), MAE by 75% ($0.1269 \rightarrow 0.0317$), and MVR-Wait from 62.5% to 30.2% on the 64-state grid for Qwen2.5-72B-Instruct; DeepSeek V4 saw MSE drop from 0.0379 to 0.0024 (−94%) and MVR-Wait from 34.4% to 7.3%; MiroThinker-1.7-mini saw MSE drop from 0.0872 to 0.0054 (−94%) and MVR-Wait from 10.4% to 8.3%. All three EFR predictions achieve $\rho \geq 0.69$, exceeding the construction-only null distribution ($\rho = 0.367$, 95% CI $[0.147, 0.566]$) by a wide margin. On the LLM-matched held-out subset ($N = 813$ *alternative-level binary-outcome rows* from $N = 205$ held-out respondents (each row is one of three alternatives within one of six tasks; the held-out pool contains $N = 3,690$ alt-rows; after excluding one anomalous wait=2 alt-row from a single respondent, 3,689 rows remain under the intended design, of which 813 are matched and 2,876 are unmatched; the matched subset is the literal overlap between the intended DCE design and the LLM grid on the three focal attributes (waiting time, efficacy, and side-effect burden), namely $\text{wait} \in \{0, 6\}$, $\text{eff} \in \{0.5, 0.7\}$, $\text{se} \in \{1, 2, 3\}$, populating 11 of 12 candidate cells), with cash and origin held at the grid reference values (cash = 0, origin = 0); see Supplementary Material Table S12 (attribute-level mapping) for the full attribute-level mapping)), EFR log loss (0.699) was lower than unconstrained LLM (0.827) and Brier score (0.252 vs. 0.314), with the largest gains at high-confidence LLM signals.

Table 2: Behavioral consistency under out-of-design policy scenarios.

Model	Method	MVR ($\downarrow$)	MAD ($\downarrow$)	Spearman ρ ($\uparrow$)
<i>Panel A: Core scenarios (6 paired policy descriptions)</i>				
Qwen2.5-72B-Instruct	Unconstrained LLM	0.333 (2/6)	0.290	0.12
Qwen2.5-72B-Instruct	BDT	0.000	0.072	0.87
DeepSeek V4	Unconstrained LLM	0.667 (4/6)	0.257	-0.07
DeepSeek V4	BDT	0.000	0.064	0.87
<i>Panel B: Extended scenarios (14 paired; MVR-Wait defined only for 2 wait-varying pairs)</i>				
Qwen2.5-72B-Instruct	Unconstrained LLM	0.000 (0/2)	0.125	-0.01
Qwen2.5-72B-Instruct	BDT	0.000	0.031	0.93
DeepSeek V4	Unconstrained LLM	0.000 (0/2)	0.209	-0.06
DeepSeek V4	BDT	0.000	0.052	0.88

Note. Twenty paired scenarios (40 narrative descriptions total); eight vary waiting time and twelve vary only narrative content. MVR-Wait is reported on the eight wait-varying pairs (Panel A: 6; Panel B: 2); the twelve narrative-only pairs are reported as MAD/ ρ only. MAD is mean absolute deviation from P_{emp} . *Panel B Spearman ρ is reported alongside Kendall τ_b to address tied P_{emp} :* with only six distinct P_{emp} across 28 scenarios, a pure-noise LLM null attains $\rho \approx 0.89$ (Appendix B of the Supplementary Material), placing the reported $\rho \in [0.87, 0.93]$ within the structural noise floor. These are descriptive robustness measures, not population-level estimates.

4.3 Out-of-Design Policy Scenario Evaluation

The evaluations in Section 4.2 were conducted on a structured 64-state grid defined over the DCE attribute dimensions. We next examine whether empirical-frontier regularization preserves behavioral consistency when LLMs are applied to richer policy narratives that extend beyond the original DCE design. Unlike the previous experiments, these scenarios require the LLM to interpret unstructured textual descriptions while maintaining the waiting-time monotonicity implied by the empirical behavioral reference.

Twenty paired policy scenarios (40 narrative descriptions total) were constructed. Panel A (six pairs, A–F) varies waiting time (2 vs. 6 months; all DCE and evaluation-grid waiting times in this paper are reported in months, with walk-in appointment coded as 0 months) and includes access-related descriptions (walk-in, online registration, paperwork). Panel B (fourteen pairs, G–T) varies waiting time in only two pairs (I, J); the other twelve hold waiting time constant and vary side-effect, efficacy, or fictional narrative content (VIP access, cash rewards). The MVR-Wait metric is therefore defined only for the eight wait-varying pairs; the twelve narrative-only pairs test LLM sensitivity to non-wait attribute text.

Table 2 summarizes the out-of-design evaluation. In the core scenarios (Panel A), the unconstrained LLMs violated the expected waiting-time ordering in 2 of 6 paired scenarios for Qwen2.5-72B-Instruct and 4 of 6 paired scenarios for DeepSeek V4. After empirical-frontier regularization, no monotonicity violations were observed for either model, and rank agreement with the empirical behavioral reference increased substantially.

The differences became more pronounced in the extended scenarios (Panel B). Rich textual narratives containing fictional incentives and complex policy descriptions reduced the rank agreement of the unconstrained LLMs to approximately zero for both models. Among the two wait-varying pairs in Panel B, BDT maintained zero monotonicity violations (0/2); the twelve narrative-only pairs are reported as MAD and ρ only (all 20 paired scenarios = 40 narrative descriptions; only Qwen2.5-72B-Instruct and DeepSeek V4 were evaluated on the OOD panel, MiroThinker-1.7-mini was not). Rank agreement with the empirical reference was $\rho = 0.93$ for Qwen2.5-72B-Instruct

and $\rho = 0.88$ for DeepSeek V4. These out-of-design results characterize structural consistency of the anchored outputs with the extrapolated empirical reference; they do not establish predictive validity for human behavior outside the observed DCE design.

This experiment evaluates final anchored predictions; the separate contributions of language understanding and empirical regularization are not isolated here.

4.4 Policy Ranking Consistency

EFR increased agreement with the Pure-DCE policy ranking (a structural diagnostic, not an independent validation): Qwen2.5-72B-Instruct Spearman PRC $0.333 \rightarrow 0.867$; DeepSeek V4 $0.133 \rightarrow 0.600$; MiroThinker-1.7-mini $0.533 \rightarrow 0.800$. Because the Static-BDT blend contains 75% P_{emp} by construction, increased agreement with the Pure-DCE ranking is a mechanical consequence of the convex anchor, not an independent behavioral validation. PRC is therefore reported in the supplement as a structural diagnostic only; no significance test is performed on the 15 pairwise comparisons because they share six underlying policy effects and are not 15 independent Bernoulli trials.

5 Discussion

5.1 Principal Findings

EFR improves behavioral consistency of LLM-based synthetic agents without retraining. Three findings emerge. *First*, unconstrained LLMs produce biased choice probabilities: waiting-time monotonicity violations ranged 10.4–62.5% on the 64-state grid (Qwen2.5-72B-Instruct: 62.5%; DeepSeek V4: 34.4%; MiroThinker-1.7-mini: 10.4%); rank correlation with the empirical frontier was weakly positive (Qwen2.5-72B-Instruct: $\rho = 0.061$; DeepSeek V4: 0.302; MiroThinker-1.7-mini: 0.531). *Second*, EFR reduced these violations to 7.3–30.2% across the three LLMs (Qwen2.5-72B-Instruct: 30.2%; DeepSeek V4: 7.3%; MiroThinker-1.7-mini: 8.3%) and raised frontier-alignment ρ to ≥ 0.69 (Qwen2.5-72B-Instruct: 0.689; DeepSeek V4: 0.812; MiroThinker-1.7-mini: 0.756). Held-out DCE log loss and Brier score (Static-BDT: 0.699 / 0.252; Pure-DCE: 0.679 / 0.242; unconstrained LLM: 0.827 / 0.314 on the matched held-out subset, $N = 813$ *alternative-level binary-outcome rows* from $N = 205$ held-out respondents, i.e. the 813 rows in the LLM-matched subset of the held-out pool (the held-out pool contains 3,690 alt-rows; after excluding one anomalous wait=2 alt-row, 3,689 rows remain under the intended design, of which 813 are matched and 2,876 are unmatched) are reported as sensitivity diagnostics; the $\lambda = 0.25$ main specification does not beat the Pure-DCE endpoint on these held-out metrics and is not selected on them. *Third*, agreement with the Pure-DCE policy ranking increased after EFR (Qwen2.5-72B-Instruct Spearman PRC $0.333 \rightarrow 0.867$; MiroThinker-1.7-mini $0.533 \rightarrow 0.800$); this is a structural diagnostic of the 75% P_{emp} blend rather than an independent behavioral validation, and is reported in the supplement without significance testing.

5.2 Methodological Contributions

Our primary contribution is *inference-time empirical regularization*, a procedure that aligns LLM decisions with empirically estimated behavioral structure without modifying model weights. The design reflects theory-informed machine learning: established empirical models provide structural priors for generative AI systems. EFR is a wrapper, not a model: it can be applied to any LLM

(closed-source or open-source) and any empirical choice frontier (mixed logit, latent class, or non-parametric). The convex combination $\pi_{\text{EFR}} = \lambda\pi_{\text{LLM}} + (1 - \lambda)P_{\text{emp}}$ is the simplest possible anchor; future work could explore non-convex anchors (e.g., mixture-of-experts gating, Bayesian shrinkage toward P_{emp} priors).

A robustness stress test on eight Panel B scenarios across three smaller open-source models (4,800 prompts) corroborated this view: Qwen2.5-7B produced constant outputs (zero variance, default-answer signature), DeepSeek V4 (Flash variant) achieved 0% parse on the waiver-language N1 scenario. *These failures are a safety property, not evidence of LLM value-add:* when LLM output is degenerate, EFR continues to assign 75% of the weight to P_{emp} , bounding—but not eliminating—the LLM component (EFR is not a reliability gate, and the $\lambda = 0.25$ weight is fixed rather than adaptive). Our use of LLMs as anchored narrative components differs from prior tool-oriented LLM work: the empirical frontier, not the LLM, supplies the behavioral signal; the LLM component is permitted only in the convex blend with P_{emp} .

5.3 Implications for Health Policy Analysis

The EFR framework has three HTA implications. *First*, a recent DCE unlocks a low-marginal-cost narrative-policy simulation pipeline: HTA agencies need only the LLM API, DCE estimates, and a deterministic blend. This is most valuable for early-stage policy screening (e.g., new vaccine reimbursement tiers) where new DCE cost is prohibitive. *Second*, EFR extends the productive life of an existing DCE: EFR applies to narrative policy descriptions (fictional incentives, side-effect profiles, communication strategies) outside the original 4–6 attribute factorial grid. *Third*, the framework is best suited to early-stage screening, not high-stakes final decisions; for the latter, EFR outputs should be triangulated with traditional stated-preference methods.

Pre-empted limitations. We highlight four design choices that, in our judgment, deserve explicit acknowledgement before external review. *(i) Wuhan DCE, single-jurisdiction sample.* The empirical frontier is estimated from one DCE fielded in a single Chinese city (collection period and full DCE questionnaire reported in the public code repository at <https://github.com/REDACTED/behavioral-digital-twins>); preference stability across pandemic phases is not established. Replication across additional jurisdictions and time periods is planned future work (Conclusion). *(ii) Conditional-logit specification and frontier stability.* The static frontier uses a 6-parameter grouped/conditional logit (five attribute coefficients plus an opt-out alternative-specific constant) on the full DCE sample, not a random-coefficients mixed logit. Respondent-cluster bootstrap (2,000 replicates) confirms frontier stability: Spearman ρ between bootstrap and canonical frontier has mean 0.988 (median 0.993; 95% CI [0.948, 1.000]) and all six coefficients retain their sign in $\geq 97.7\%$ of replicates (Appendix J of the Supplementary Material). The corrected opt-out ASC narrows P_{emp} to [0.47, 0.64]; we trade within-DCE-design contrast for a behaviorally honest specification. *(iii) $\lambda = 0.25$ as a conservative design choice.* The blending weight is a fixed conservative design specification preserving a non-zero LLM contribution while keeping the empirical DCE dominant; it is not a test-set-optimized predictive optimum. The held-out DCE log loss and Brier score are reported as sensitivity diagnostics and were not used to select λ . Structurally, this weight retains only one quarter of the raw LLM–frontier deviation pointwise, mechanically bounding frontier-relative squared deviation to $\lambda^2 = 6.25\%$ of the unconstrained LLM value (Supplementary Material F.2). *(iv) Inference-time regularization vs. retraining.* EFR is a convex combination of two streams, not a learned gate; non-convex anchors (mixture-of-experts gating, Bayesian shrinkage toward P_{emp} priors) remain future work.

5.4 Why Not Use the Discrete Choice Model Alone?

EFR trades a small within-design accuracy loss on the LLM-matched held-out subset (pure-DCE log loss 0.679; EFR 0.699; $\Delta = 0.020$) for narrative-policy evaluation. The NDS pilot ($n = 2$) corroborates: isotonic NDS $\rho = -0.943$ vs. EFR $\rho = +0.894$ for DeepSeek V4. **Figure 1** reports the held-out human-DCE predictive performance (log loss and Brier score) of the Pure-DCE, Static-BDT Anchor (EFR, $\lambda = 0.25$), and Raw LLM (Qwen2.5-72B-Instruct) endpoints, with respondent-cluster bootstrap 95% confidence intervals. EFR does not outperform the Pure-DCE endpoint; it recovers much of the DCE’s predictive discipline while retaining a nonzero narrative component (EFR log loss 0.699 vs. Pure-DCE 0.679, $\Delta = 0.020$). The OOD panel-level diagnostics (Spearman ρ and MAD on the 20 paired narrative scenarios) are reported as supplementary structural diagnostics in Table 2; those diagnostics fall inside the construction-only null band and are *not* reported as independent behavioral validation.

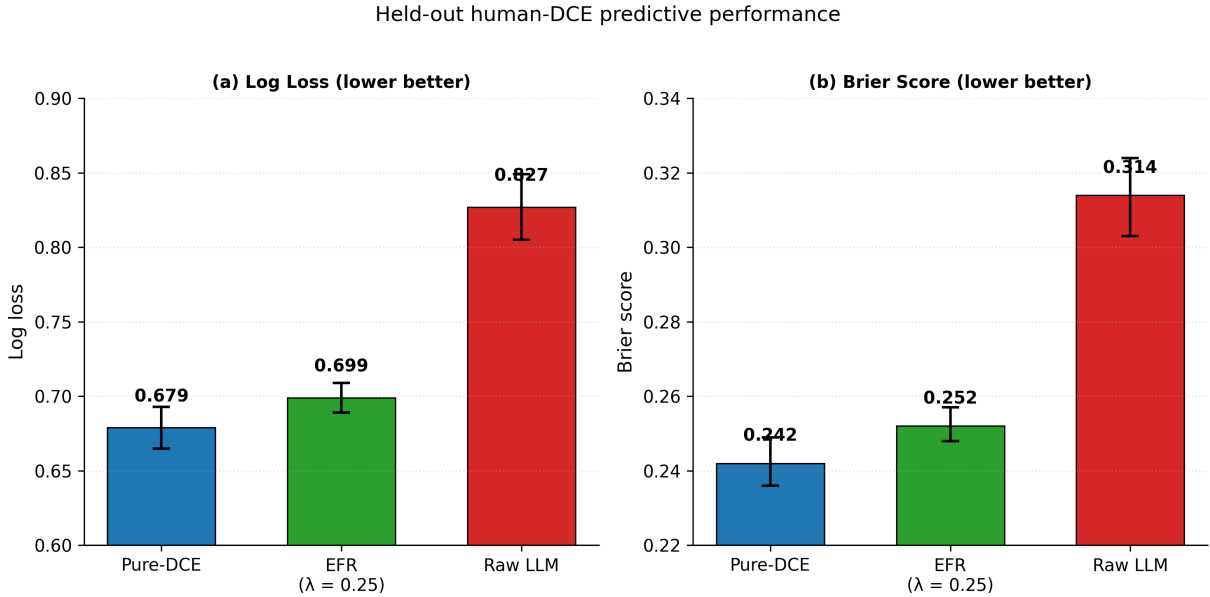


Figure 1: Held-out human-DCE predictive performance on the $N = 813$ alternative-level binary-outcome rows from the $N = 205$ held-out respondents (each row is one of three alternatives within one of six tasks), with respondent-cluster bootstrap 95% confidence intervals (2,000 replicates). (a) Log loss; (b) Brier score. *EFR does not outperform the Pure-DCE endpoint; it recovers much of its predictive discipline (EFR log loss 0.699 vs. Pure-DCE 0.679, $\Delta = 0.020$) while retaining a nonzero narrative component permitted inside the convex blend with P_{emp} . The Raw LLM endpoint is materially worse on both metrics (log loss 0.827, Brier 0.314).*

5.5 Implementation and Policy Recommendations

Significance. EFR enables LLM-based behavioral simulation that is auditable and behaviorally grounded by anchoring LLM outputs to an existing DCE’s empirical frontier. The hybrid inherits the LLM’s flexibility for unstructured policy scenarios and the DCE’s reliability for established preference patterns, with the fixed $\lambda = 0.25$ anchor bounding, rather than eliminating, the influence of unreliable LLM output; EFR is not a reliability gate and λ is not adaptive. Implementation requires no model retraining: an analyst estimates a 6-parameter grouped/conditional logit with

five attribute coefficients and an opt-out alternative-specific constant from an existing DCE [12, 7], generates LLM predictions, and blends them through $\pi_{\text{ad}} = \lambda\pi_{\text{LLM}} + (1 - \lambda)P_{\text{emp}}$ ($\lambda = 0.25$). Three policy implications follow. *With a DCE*, static-BDT provides a path from evidence to narrative policy simulation. *Without a DCE*, the framework is not a substitute for stated-preference data. *Best for early-stage screening*, not high-stakes final decisions.

5.6 Conclusion

EFR anchors LLM outputs to DCE estimates without retraining. Held-out DCE log loss and Brier score (Static-BDT: 0.699 / 0.252; Pure-DCE: 0.679 / 0.242; unconstrained LLM: 0.827 / 0.314 on the matched held-out subset, $N = 813$ *alternative-level binary-outcome rows* from $N = 205$ held-out respondents, i.e. the 813 rows in the LLM-matched subset of the held-out pool (the held-out pool contains 3,690 alt-rows; after excluding one anomalous wait=2 alt-row, 3,689 rows remain under the intended design, of which 813 are matched and 2,876 are unmatched) are reported as sensitivity diagnostics; the 75% weight on P_{emp} bounds, rather than eliminates, the influence of unreliable LLM output; EFR is not a reliability gate and λ is not adaptive, so the bound is fixed rather than adaptive. On the structured 64-state grid, MVR-Wait fell from 10.4–62.5% to 7.3–30.2% across three LLMs (Qwen2.5-72B-Instruct: 62.5% $\rightarrow$ 30.2%; DeepSeek V4: 34.4% $\rightarrow$ 7.3%; MiroThinker-1.7-mini: 10.4% $\rightarrow$ 8.3%) under EFR anchoring at $\lambda = 0.25$ (Table 1). *In the eight wait-varying OOD narrative pairs (40 narrative descriptions total, evaluated with Qwen2.5-72B-Instruct and DeepSeek V4 only), EFR eliminated the observed waiting-time reversals on Panel A (Qwen: 2/6 to 0/6; DeepSeek: 4/6 to 0/6) and held the Panel B wait-varying pairs at 0/2 for both models; the twelve narrative-only pairs (varying side-effect, efficacy, or fictional narrative content while holding wait constant) are reported as MAD and Spearman ρ only.* EFR is best understood as a constrained narrative-extension mechanism for an existing DCE, with the LLM component permitted only inside the convex blend with P_{emp} . The empirical reference, not the LLM, supplies the behavioral signal; whether the narrative extension is behaviorally more accurate on out-of-design scenarios remains to be established by future work with human ground truth. Three directions are planned future work: (i) replication across additional diseases, jurisdictions, and policy domains; (ii) exploration of non-convex anchoring mechanisms (e.g., mixture-of-experts gating); (iii) quantification of predictive value in actual HTA decisions where the LLM contributes incremental information.

References

- [1] Gati Aher, Rosa I. Arriaga, and Adam Tauman Kalai. Using large language models to simulate multiple humans and replicate human subject studies, 2022.
- [2] Lisa P. Argyle, Ethan C. Busby, Nancy Fulda, Joshua Gubler, Christopher Rytting, and David Wingate. Out of one, many: Using language models to simulate human samples. *Political Analysis*, 31(3):337–351, 2023.
- [3] Rishi Bommasani, Drew A. Hudson, Ehsan Adeli, Russ Altman, Simran Arora, Sydney von Arx, Michael S. Bernstein, Jeannette Bohg, Antoine Bosselut, Emma Brunskill, et al. On the opportunities and risks of foundation models. *arXiv preprint arXiv:2108.07258*, 2021.
- [4] John F. P. Bridges, A. Brett Hauber, Deborah Marshall, Andrew Lloyd, Lisa A. Prosser, Dean A. Regier, F. Reed Johnson, and Josephine Mauskopf. Conjoint analysis applications in health—How are studies being designed and reported?: An update: Current practices in ISPOR conjoint analysis applications. *Value in Health*, 16(2):287–298, 2013. doi: 10.1016/j.jval.2012.12.012.
- [5] Amy Hai Yan Chan et al. Vaccine decision making in new zealand: A discrete choice experiment. *BMC Public Health*, 24:447, 2024. doi: 10.1186/s12889-024-17965-5.
- [6] Lucinda Gullison and Feng Fu. Working with large language models to enhance messaging effectiveness for vaccine confidence. *arXiv preprint arXiv:2504.09857*, 2025.
- [7] A. Brett Hauber, Juan Marcos González, Catharina G. M. Groothuis-Oudshoorn, Thomas Prior, Deborah A. Marshall, Charles Cunningham, Maarten J. IJzerman, and John F. P. Bridges. Statistical methods for the analysis of discrete choice experiments: A report of the ISPOR conjoint analysis good research practices task force. *Value in Health*, 19(4):300–315, 2016. doi: 10.1016/j.jval.2016.04.004.
- [8] David A. Hensher and William H. Greene. The mixed logit model: The state of practice. *Transportation*, 30(2):133–176, 2003.
- [9] David A. Hensher, John M. Rose, and William H. Greene. *Applied Choice Analysis: A Primer*. Cambridge University Press, Cambridge, 2nd edition, 2015.
- [10] Arend Hintze, Charu Bisht, Jory Schossau, and Ralph Hertwig. Nonlinear transformation of probabilities by large language models. *Computers in Human Behavior: Artificial Humans*, 6: 100227, 2025. doi: 10.1016/j.chbah.2025.100227.
- [11] John J. Horton. Large language models as simulated economic agents: What can we learn from homo silicus? *NBER Working Paper*, (w31122), 2023. doi: 10.3386/w31122.
- [12] F. Reed Johnson, Esther Lancsar, Deborah Marshall, Vikram Kilambi, Andreas Muehlbacher, Dean A. Regier, Brian W. Bresnahan, Brett J. Kanninen, and John F. P. Bridges. Constructing experimental designs for discrete-choice experiments: Report of the ISPOR conjoint analysis good research practices task force. *Value in Health*, 16(1):3–13, 2013. doi: 10.1016/j.jval.2012.08.223.

- [13] Fatima Koaik, Aayush Gupta, and Farahan Raza Sheikh. Llm powered social digital twins: A framework for simulating population behavioral response to policy interventions. *arXiv preprint arXiv:2601.06111 [cs.AI]*, January 2026. 13 pages, 1 figure, 4 tables; available at <https://arxiv.org/abs/2601.06111>.
- [14] Yiming Li and Dacheng Tao. Position: AI agents are not (yet) a panacea for social simulation, February 2026.
- [15] Jordan J. Louviere, David A. Hensher, and Joffre D. Swait. *Stated Choice Methods: Analyzing and Interpreting Data*. Cambridge University Press, Cambridge, 2000.
- [16] Nicholas Luo, Stephanie Tan, Esha Wadhwa, and Jiraganya Pwu. Simulating sponsor-HTA conversations: A role-based multi-agent large language model framework. *Value in Health*, 28(12):S1–S6, 2025. Conference abstract P4; multi-agent LLM framework for sponsor-HTA interactions.
- [17] Xinyi Mu, Xuanwen Ding, Qi He, Liang Wang, Jingcong Liang, Xinnong Zhang, Libo Sun, Jiayu Lin, Jie Zhou, Xuanjing Huang, and Zhongyu Wei. From individual to society: A survey on social simulation driven by large language model-based agents. *arXiv preprint arXiv:2412.03563*, 2024.
- [18] Joon Sung Park, Joseph O’Brien, Carrie J. Cai, Meredith Ringel Morris, Percy Liang, and Michael S. Bernstein. Generative agents: Interactive simulacra of human behavior. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*, 2023. doi: 10.1145/3586183.3606763.
- [19] Shibani Santurkar, Esin Durmus, Faisal Ladhak, Cinoo Lee, Percy Liang, and Tatsunori Hashimoto. Whose opinions do language models reflect? In *International Conference on Machine Learning (ICML)*, 2023.
- [20] Neil K. R. Sehgal, Sunny Rai, Manuel Tonneau, Anish K. Agarwal, Joseph Cappella, Melanie Kornides, Lyle Ungar, Alison Buttenheim, and Sharath Chandra Guntuku. Conversations with ai chatbots increase short-term vaccine intentions but do not outperform standard public health messaging. *arXiv preprint arXiv:2504.20519*, 2025.
- [21] Akshay Shree, Oliver Phillips, Ryan Sullivan, Pablo Garcia, and Aarti Bansal. Stability of a large language model for data extraction in systematic reviews: A sensitivity analysis. *Value in Health*, 28(12):1832–1839, 2025. Conference abstract MSR105; full sensitivity analysis by varying time of day and geographic location.
- [22] Vikas Soekhai, Esther W. de Bekker-Grob, Alan R. Ellis, and Caroline M. Vass. Discrete choice experiments in health economics: Past, present and future. *PharmacoEconomics*, 37(2):201–226, 2019. doi: 10.1007/s40273-018-0734-2.
- [23] Petter Törnberg and Marlon Larooij. Validation is the central challenge for generative social simulation: A critical review of LLMs in agent-based modeling. *Artificial Intelligence Review*, 59(1):15, 2025. doi: 10.1007/s10462-025-11412-6.
- [24] Olivier Toubia, Junchao Yang, Deepak Pai, Youngseok Chae, and Jonah Berger. Twin-2K-500: A data set for building digital twins of US adult consumers from their Amazon purchase histories. *Marketing Science*, 2025. Forthcoming; preprint cited via DOI 10.1287/mksc.2025.0262.

- [25] Kenneth E. Train. *Discrete Choice Methods with Simulation*. Cambridge University Press, 2 edition, 2009.
- [26] Emma Von Hoene, Amira Roess, Hamdi Kavak, and Taylor Anderson. Synthetic population generation with public health characteristics for spatial agent-based models. *PLOS Computational Biology*, 2025. doi: 10.1371/journal.pcbi.1012439.
- [27] Zhengyu Wang, Tomasz Pawlak, and Konstantinos Sivakumar. Heating-choice stated preferences with large language models. *Working Paper*, 2025.
- [28] Zhengyu Wang et al. Can LLMs simulate human responses? *arXiv preprint arXiv:2503.10652*, 2025.

Supplementary Materials

The following supplementary materials accompany this manuscript and are available in the online appendix.

A. Tables

- **Table S1.** LLM decoding and endpoint parameters used in the main, supplementary, and stress-test evaluations.
- **Table S2.** *Administered* DCE attribute levels: waiting time $\{0, 1, 3, 6\}$ months (4 real design levels; the raw coded data contain a single 1-row outlier at wait = 2 months from one respondent which we exclude as a data anomaly because it does not represent an administered design level), vaccine efficacy $\{0, 50, 70, 95\}\%$, side-effect burden $\{0, 10, 20, 30\}\%$, cash incentive $\{50, 200, 800\}$ RMB, vaccine origin $\{\text{Imported, Domestic}\}$. *Note: 0 months (walk-in appointment), 0% vaccine efficacy, 0% side effects, and 0 RMB cash incentive appear in the empirical data only as the opt-out alternative (the 6 administered choice tasks each present two non-opt-out profiles plus an opt-out alternative that is not part of the non-opt-out DCE design matrix).* The 12 administered non-opt-out attribute combinations per respondent (1 unique design block shared by all 1027 respondents) enumerate 12 of the $3 \times 3 \times 3 \times 2 \times 4 = 216$ non-opt-out candidate cells (5.6% coverage).
- **Table S3.** Attribute coding dictionary: DCE questionnaire display, estimation coding, 64-state evaluation-grid coding, and coefficient unit definitions.
- **Table S4.** Conditional-logit estimates (corrected 6-parameter specification) used to construct the empirical choice frontier. Standard errors from BFGS Hessian; bootstrap 95% confidence intervals for the 6 coefficients.
- **Table S5.** λ -sensitivity for frontier-alignment diagnostics on the 64-state grid. Pure-DCE baseline, unconstrained LLM, and Static-BDT Anchor evaluated across $\lambda \in \{0.00, 0.10, 0.20, 0.25, 0.30, 0.40, 0.50\}$.
- **Table S6.** λ -sensitivity for held-out DCE predictive validation on the 813 *alternative-level* rows subset (from 205 held-out respondents). Alt-level binary log loss, Brier score, and AUC across λ .
- **Table S7.** Data-scaling ablation: Static-BDT Anchor ($\lambda = 0.25$) vs. unconstrained LLM across training fractions (30%, 50%, 70%, 100%).
- **Table S8.** Parse success rates across models and experiment scales.
- **Table S9.** Runtime summary for full-scale unconstrained LLM generation.
- **Table S10.** Policy Ranking Consistency (PRC) relative to the Pure-DCE reference; descriptive structural diagnostic, no significance test.
- **Table S11.** Bootstrap intervals for selected frontier-alignment diagnostics.
- **Table S12.** Attribute-level mapping between the administered DCE design and the 64-state LLM evaluation grid. The matched subset is restricted to cells where the DCE design level and the grid level coincide as a `float` value (no rescaling, no interpolation).
- **Table S13.** Covariate balance between matched and unmatched held-out subsets.
- **Table S14.** Subgroup heterogeneity analysis of Static-BDT ($\lambda = 0.25$) on the 813 *alternative-level* rows subset (from 205 held-out respondents).
- **Table S15.** Task-level multinomial held-out log loss on the 2-alternative subset (A vs. B, opt-out excluded).

- **Table S16.** Task-level multinomial held-out log loss on the 3-alternative full tasks (A, B, C).

B. Figures

- **Figure S1.** Data-scaling ablation: Static-BDT Anchor ($\lambda = 0.25$) evaluated across DCE training shares (30%, 50%, 70%, 100%); companion to the data-scaling narrative in Appendix G.

C. Replication Materials

All data and code are publicly available at <https://github.com/REDACTED/behavioral-digital-twins>, including: 240-row OOD probability dataset, 64-state grid, 20 OOD paired scenarios (40 narrative descriptions), NDS baseline script, bootstrap CI computations, and subgroup heterogeneity analysis.